

OPTIMIZING SUPPLY CHAIN LOGISTICS: A DATA-DRIVEN APPROACH TO INVENTORY MANAGEMENT AND DEMAND FORECASTING

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DECLARATION

I hereby declare that the work presented in this project report, titled “**Optimizing Supply Chain Logistics: A Data-Driven Approach to Inventory Management and Demand Forecasting**,” is my own work completed as part of the Data Science and AI Summer Training Program by India Data Research Academy (IDRA). The data analysis, preprocessing, feature engineering, statistical analysis, machine learning implementation, evaluation, and interpretation presented in this report were carried out as part of this project. All external sources, references, and supporting materials used in the preparation of this report have been appropriately acknowledged.

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ABSTRACT

Effective inventory management and demand forecasting are essential for reducing stock-related risks and improving supply chain planning. This project presents a data-driven approach to analyzing supply chain demand and forecasting next-day units sold using product, warehouse, supplier, inventory, pricing, and promotional information. The study uses a dataset containing 91,250 observations and 15 variables representing daily supply chain operations. The analysis involved data cleaning, exploratory data analysis, statistical testing, feature engineering, and machine learning. Historical demand features such as lagged demand and rolling averages, along with temporal, operational, commercial, and categorical features, were used to predict next-day demand. A chronological 80:20 train-test split was applied to evaluate model performance on future observations. Persistence Baseline, Linear Regression, Decision Tree, and Random Forest models were compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 . Linear Regression achieved the best test performance among the evaluated models, with an MAE of 5.4290, RMSE of 6.8054, and R^2 of -0.2733. Statistical analysis also identified a significant association between promotional activity and demand. Overall, the study provides a baseline framework for data-driven demand analysis while highlighting the need for richer data and improved forecasting methods for stronger predictive performance.

Keywords: Supply Chain Management; Demand Forecasting; Inventory Management; Machine Learning; Regression Analysis

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1. INTRODUCTION

1.1 Background

Supply chain management plays an important role in ensuring that products are available at the right time and in the required quantity. Effective inventory management depends on understanding product demand, monitoring inventory levels, coordinating with suppliers, and responding to changes in sales patterns. Inaccurate demand estimation can result in excess inventory, unnecessary holding costs, or insufficient stock to meet customer requirements.

Data-driven approaches provide an opportunity to analyze historical supply chain information and identify patterns that can support inventory and demand planning. Machine learning techniques can use historical demand, temporal patterns, operational information, pricing, and promotional activity to estimate future demand. Such analysis can help organizations make more informed decisions regarding replenishment and inventory planning.

This project focuses on analysing supply chain data and developing a machine learning approach for next-day demand forecasting, with units sold used as the primary measure of demand.

1.2 Context of the Problem

Supply chain decisions involve multiple factors, including product demand, warehouse operations, supplier characteristics, inventory levels, pricing, and promotional activities. Demand can vary over time, making it difficult to determine appropriate inventory and replenishment levels using fixed assumptions alone.

In this context, supply chain planners and inventory managers can benefit from analytical methods that identify demand patterns and provide estimates of future sales. This project uses SKU-level and warehouse-level supply chain observations to examine demand behaviour and evaluate whether historical and operational information can be used to predict the number of units sold on the following day.

The analysis also examines the relationship between promotional activity and demand, along with relationships between demand and selected operational and commercial variables. The findings are intended to provide analytical insights that can support demand planning and inventory management.

1.3 Scope of the Study

The scope of this study covers the analysis of the provided supply chain dataset and the development of a one-day-ahead demand forecasting model. The study considers product, warehouse, supplier, regional, inventory, operational, pricing, and promotional information. It includes data cleaning, exploratory data analysis, statistical analysis, historical and temporal feature engineering, machine learning model development, model evaluation, and interpretation.

The forecasting task predicts `Next_Day_Units_Sold` using information available from the current and previous observations. Persistence Baseline, Linear Regression, Decision Tree, and Random Forest models are evaluated using a chronological train-test split. Model performance is assessed using MAE, RMSE, and R^2 .

The study does not attempt to provide a complete automated inventory optimization system or establish causal relationships between supply chain variables and demand. The findings and model performance are specific to the characteristics and time period of the provided dataset.

2. PROBLEM STATEMENT AND MOTIVATION

Problem Statement: How can historical demand, inventory, operational, pricing, promotional, and supply-related information be used to predict next-day units sold and support data-driven inventory management?

Supply chain operations require organizations to maintain sufficient inventory while avoiding unnecessary stock accumulation. Demand fluctuations can make inventory and replenishment decisions difficult, particularly when decisions are based on fixed assumptions or limited historical analysis. A reliable understanding of demand patterns can help supply chain planners make more informed decisions about inventory levels and replenishment requirements.

The motivation for this project is to investigate whether the available supply chain data can provide useful information for understanding demand patterns and forecasting future sales. The project applies exploratory and statistical analysis to identify important relationships in the data and evaluates machine learning models for predicting next-day demand. The resulting analysis can provide a data-driven baseline for demand planning and inventory management while also identifying areas where additional data and improved forecasting methods may be required.

3. OBJECTIVES

- To analyze the selected supply chain dataset and identify patterns in demand, inventory, pricing, promotional activity, and operational variables.
- To examine the statistical relationships between demand and key supply chain factors, including promotional activity, inventory levels, order quantity, and regional differences.
- To engineer historical demand and temporal features for predicting next-day units sold at the SKU-warehouse level.
- To develop and compare regression-based machine learning models for next-day demand forecasting using a chronological train-test evaluation.
- To evaluate model performance using MAE, RMSE, and R^2 and translate the analytical findings into practical recommendations for demand planning and inventory management.

4. RESEARCH QUESTIONS

1. What are the key characteristics, patterns, and data quality issues present in the supply chain dataset?
2. Which demand-related, operational, pricing, and promotional variables are most strongly associated with units sold?
3. How does demand vary across different time periods, products, warehouses, and regions, and what patterns can be identified from the data?
4. How accurately can machine learning models predict next-day units sold using historical demand and available supply chain information?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The dataset used in this project is the P3 Supply Chain Dataset provided as part of the India Data Research Academy (IDRA) Data Science and AI Capstone Project. The dataset contains supply chain records covering product demand, inventory, supplier, warehouse, regional, pricing, and promotional information. The dataset was provided in CSV format and was used as the primary source for the analysis and forecasting task. The raw dataset contains 91,250 observations and 15 variables. The dataset was examined for its structure, data types, missing values, duplicate records, unique values, and descriptive statistics before performing further preprocessing and modelling.

5.2 Dataset Size and Structure

The raw dataset contains 91,250 rows and 15 columns. It includes product, warehouse, supplier, regional, demand, inventory, ordering, pricing, and promotional variables. The main demand variable is `Units_Sold`, which represents the actual units sold on a given day and is used to construct historical demand features. For the forecasting task, `Next_Day_Units_Sold` is engineered from `Units_Sold` and represents the actual units sold on the following day for the same SKU-Warehouse combination. It is therefore used as the target variable for one-day-ahead demand forecasting. A summary of the dataset size, structure, demand variable, target variable, prediction horizon, and predictor categories is presented in Table 1.

Aspect	Details
Observations	91,250
Original Variables	15
Data Format	CSV
Unit of Analysis	SKU-Warehouse-Day
Original Demand Variable	<code>Units_Sold</code>
Target Variable	<code>Next_Day_Units_Sold</code>
Prediction Horizon	1 day ahead
Predictor Categories	Historical, temporal, categorical, operational, inventory, pricing, promotional

Table 1. Summary of the Dataset

5.3 Data Dictionary

The main variables used in the analysis are summarized in Table 2, including their meaning, data type, and role in the forecasting task. The variables cover demand, product and warehouse information, inventory and supplier operations, pricing, and promotional activity. The target variable `Next_Day_Units_Sold` is engineered during preprocessing and is therefore not part of the original 15-column dataset.

Variable	Meaning	Data Type	Role
<code>Date</code>	Date of the observation	Date	Temporal Feature
<code>SKU_ID</code>	Identifies the product	Categorical	Feature
<code>Warehouse_ID</code>	Identifies the warehouse	Categorical	Feature

Supplier_ID	Identifies the supplier	Categorical	Feature
Region	Geographic region	Categorical	Feature
Units_Sold	Actual units sold on the current day	Numerical	Historical Demand
Inventory_Level	Available inventory level	Numerical	Feature
Supplier_Lead_Time_Days	Supplier delivery lead time	Numerical	Feature
Reorder_Point	Inventory level used for replenishment decisions	Numerical	Feature
Order_Quantity	Quantity ordered	Numerical	Feature
Unit_Cost	Cost of one unit	Numerical	Feature
Unit_Price	Selling price of one unit	Numerical	Feature
Promotion_Flag	Indicates whether a promotion was active	Binary	Feature
Stockout_Flag	Indicates stockout status	Binary	Excluded
Demand_Forecast	Existing demand forecast value	Numerical	Excluded
Next_Day_Units_Sold	Actual units sold on the following day	Numerical	Target

Table 2. Data Dictionary

6. DATA CLEANING

6.1 Missing Values

The raw dataset was examined for missing values across all 15 variables. The analysis found no missing values in any of the columns, with all 91,250 observations containing valid entries. Therefore, the proportion of missing values was 0% for the dataset. Since there were no missing observations, no imputation, deletion, or other missing-value treatment was required. The complete dataset was retained for subsequent data preprocessing and analysis.

6.2 Duplicate Records

The raw dataset was examined for duplicate records across all variables. The analysis identified no duplicate records among the 91,250 observations. Therefore, 0 records were removed, and the dataset size remained unchanged at 91,250 observations and 15 variables after the duplicate check. Since no duplicate observations were present, no duplicate-removal treatment was necessary before proceeding with further data cleaning and preprocessing.

6.3 Data Types and Inconsistent Values

The data types of all variables in the raw dataset were examined to ensure that they were appropriate for analysis. The dataset contains date, categorical, numerical, and binary variables. The Date variable was treated as a date field, while identifiers such as SKU_ID, Warehouse_ID, Supplier_ID, and Region were treated as categorical variables. Numerical variables such as Units_Sold, Inventory_Level, Supplier_Lead_Time_Days, Reorder_Point, Order_Quantity, Unit_Cost, and Unit_Price were retained in their appropriate numerical formats. The binary variables Promotion_Flag and Stockout_Flag were retained as binary indicators. No inconsistent categorical values or invalid entries requiring correction were identified during the data-quality checks. Therefore, no records were removed or values altered due to datatype or category inconsistencies.

6.4 Outlier Handling

Outlier analysis was performed on the numerical supply chain variables using an interquartile range (IQR)-based approach. The variables examined included Units_Sold, Inventory_Level, Supplier_Lead_Time_Days, Reorder_Point, Order_Quantity, Unit_Cost, Unit_Price, and Demand_Forecast. The IQR check identified a small number of observations outside the conventional lower and upper bounds for Units_Sold and Inventory_Level, while Order_Quantity contained a larger number of values above its upper IQR boundary because the variable has a highly concentrated distribution at zero with occasional larger order quantities.

These observations were not removed automatically, as the detected values fall within plausible ranges for supply chain operations and may represent genuine differences in daily demand, inventory levels, or order quantities rather than data-entry errors. Binary variables such as Promotion_Flag and Stockout_Flag were not treated as continuous outlier variables. Therefore, the observations were retained to preserve the original demand and operational patterns in the dataset.

6.5 Before vs After Cleaning

The results of the data-quality checks before and after cleaning are summarized in Table 3. Since the dataset contained no missing values or duplicate records and no invalid or inconsistent values requiring correction were identified, the number of observations and variables remained unchanged after the cleaning stage.

Data Quality Measure	Before Cleaning	Cleaning Action	After Cleaning
Observations	91,250	No rows removed	91,250
Variables	15	No columns removed	15
Missing Values	0	No treatment required	0
Duplicate Records	0	No duplicates removed	0

Table 3. Data Quality Before and After Cleaning

7. DATA PREPROCESSING

7.1 Categorical Encoding

The categorical variables SKU_ID, Warehouse_ID, Supplier_ID, and Region were encoded using One-Hot Encoding. The encoding was implemented using OneHotEncoder within a ColumnTransformer, with handle_unknown="ignore" to ensure that previously unseen categories in the test data could be handled without causing errors. One-hot encoding was selected because these categorical variables represent identifiers and geographic groups without an inherent numerical ordering. The encoded categorical variables were then combined with the numerical features to form the input data used by the machine learning models.

7.2 Feature Scaling and Transformation

The numerical features were not subjected to feature scaling or normalization before model training. Variables such as Inventory_Level, Supplier_Lead_Time_Days, Reorder_Point, Order_Quantity, Unit_Cost, and Unit_Price were retained in their original numerical form. The historical demand and temporal variables were created through feature engineering rather than numerical scaling. Scaling was not included in the final preprocessing pipeline, and the numerical variables were passed through unchanged. This approach kept the original units and interpretation of the supply chain variables while allowing the selected Linear Regression, Decision Tree, and Random Forest models to be evaluated using the same feature representation.

7.3 Train-Test Split

Since the project focuses on next-day demand forecasting, the data was divided using a chronological 80:20 train-test split rather than random sampling. The earlier 80% of the observations were used for training, while the later 20% were reserved for testing. The training set contains 70,000 observations, while the test set contains 17,500 observations. No random state was used because the observations were not randomly shuffled. This chronological split reflects the real-world forecasting situation in which future demand must be predicted using information available from the past. Data leakage was avoided by constructing historical lag and rolling features using only previous observations and by fitting the categorical encoding and preprocessing pipeline using the training data before transforming the test data.

7.4 Final Feature Set

After preprocessing and feature engineering, the final input feature set consisted of 21 predictor variables used for model development. These included historical demand features (lag_1, lag_7, lag_14, rolling_mean_7, and rolling_mean_14), temporal features (day_of_week, month, quarter, week_of_year, and day_of_year), categorical variables (SKU_ID, Warehouse_ID, Supplier_ID, and Region), operational and inventory variables (Inventory_Level, Supplier_Lead_Time_Days, Reorder_Point, and Order_Quantity), and commercial variables (Unit_Cost, Unit_Price, and Promotion_Flag). The target variable was Next_Day_Units_Sold, representing actual units sold on the following day. The existing Demand_Forecast and constant Stockout_Flag variables were excluded from the final feature set.

8. EXPLORATORY DATA ANALYSIS

8.1 Univariate Analysis

The distribution of the Units_Sold variable was examined to understand the overall pattern of daily demand in the dataset. Figure 1 presents the distribution of units sold across the observations.

```
# Plot the distribution of daily units sold

plt.figure(figsize=(10, 6))

plt.hist(df["Units_Sold"], bins=30)

plt.title("Distribution of Units Sold")
plt.xlabel("Units Sold")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

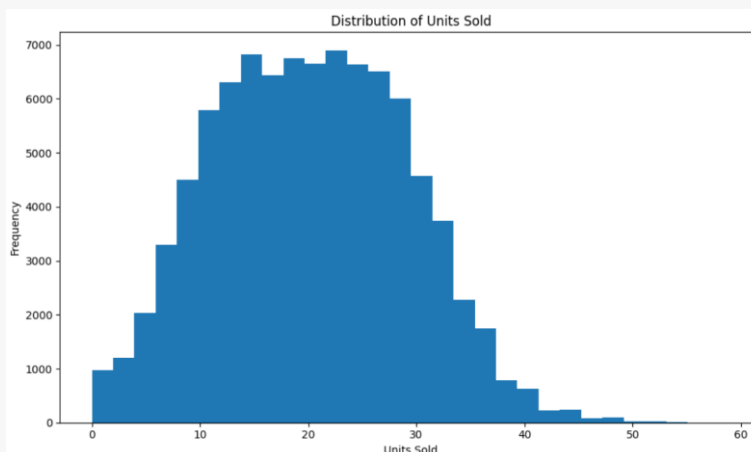


Figure 1. Distribution of Units Sold

The distribution shows that daily units sold range from 0 to 59 units, with most observations concentrated around the central range of the distribution. The distribution is broadly centered around the mean demand of approximately 20 units per observation. The presence of observations at both lower and higher demand levels indicates variation in daily sales, while the higher-demand observations represent relatively less frequent values. These variations demonstrate that demand is not constant across observations and therefore provide a basis for further analysis of temporal, promotional, and operational factors associated with demand.

8.2 Categorical Analysis

The relationship between promotional activity and demand was examined by comparing the average Units_Sold for observations with and without an active promotion. Figure 2 presents the difference in average demand between the two promotion categories.

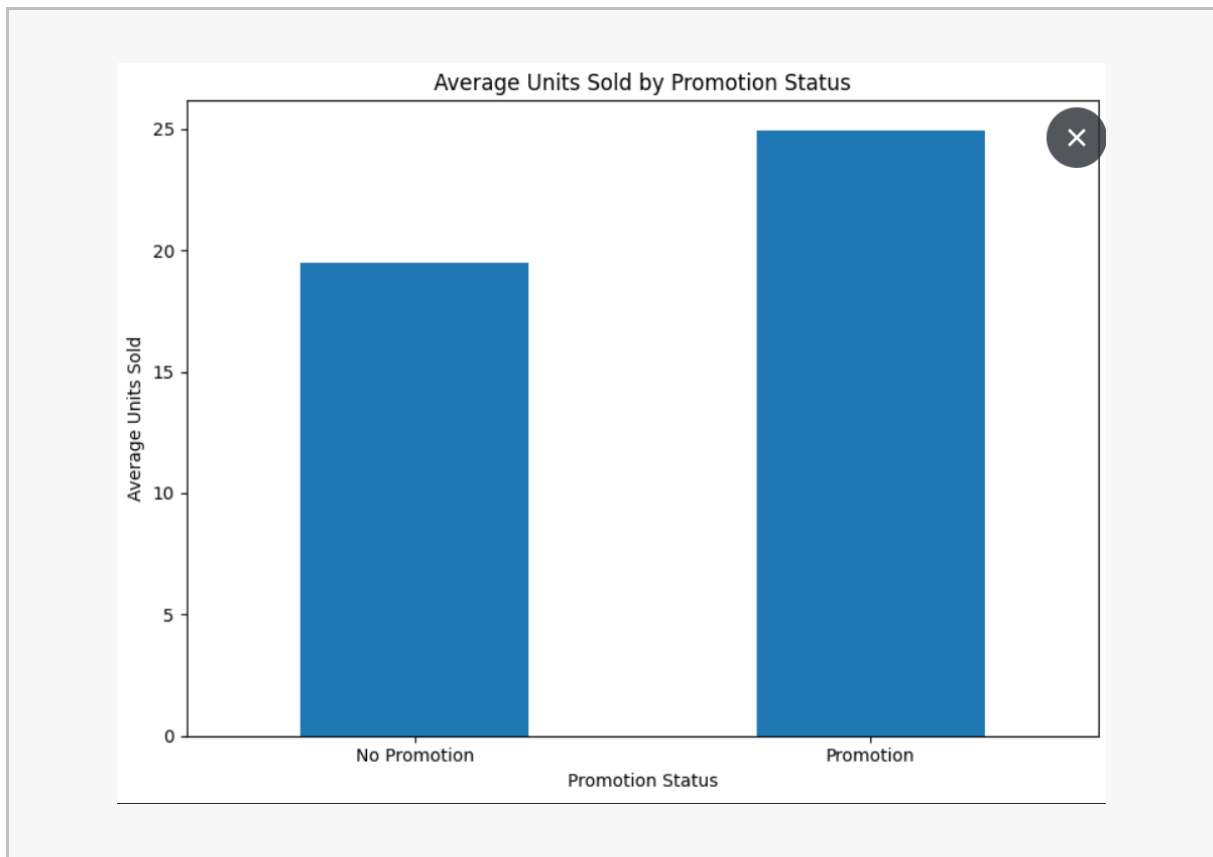
```
# Plot average demand by promotion status

plt.figure(figsize=(8, 6))

promotion_demand.plot(kind="bar")

plt.title("Average Units Sold by Promotion Status")
plt.xlabel("Promotion Status")
plt.ylabel("Average Units Sold")
plt.xticks(rotation=0)

plt.tight_layout()
plt.show()
```

**Figure 2. Average Units Sold by Promotion Status**

The figure shows a clear difference in average demand between the two promotion categories. Observations with an active promotion have an average demand of approximately 24.9 units, compared with approximately 19.5 units for observations without a promotion. This indicates that promotional periods are associated with higher observed demand in the dataset. The difference is relevant to supply chain planning because promotional activity may coincide with increased demand and therefore needs to be considered when planning inventory and replenishment requirements. However, this comparison represents an association in the observed data and does not by itself establish that promotions caused the increase in demand.

8.3 Bivariate / Relationship Analysis

The relationship between inventory availability and current-day demand was examined using Inventory_Level and Units_Sold. Figure 3 illustrates the relationship between these two variables and helps assess whether changes in inventory level are associated with differences in observed demand.

```
# Plot the relationship between inventory level and units sold

plt.figure(figsize=(10, 6))

plt.scatter(
    df["Inventory_Level"],
    df["Units_Sold"],
    alpha=0.2
)

plt.title("Inventory Level vs Units Sold")
plt.xlabel("Inventory Level")
plt.ylabel("Units Sold")

plt.tight_layout()
plt.show()
```

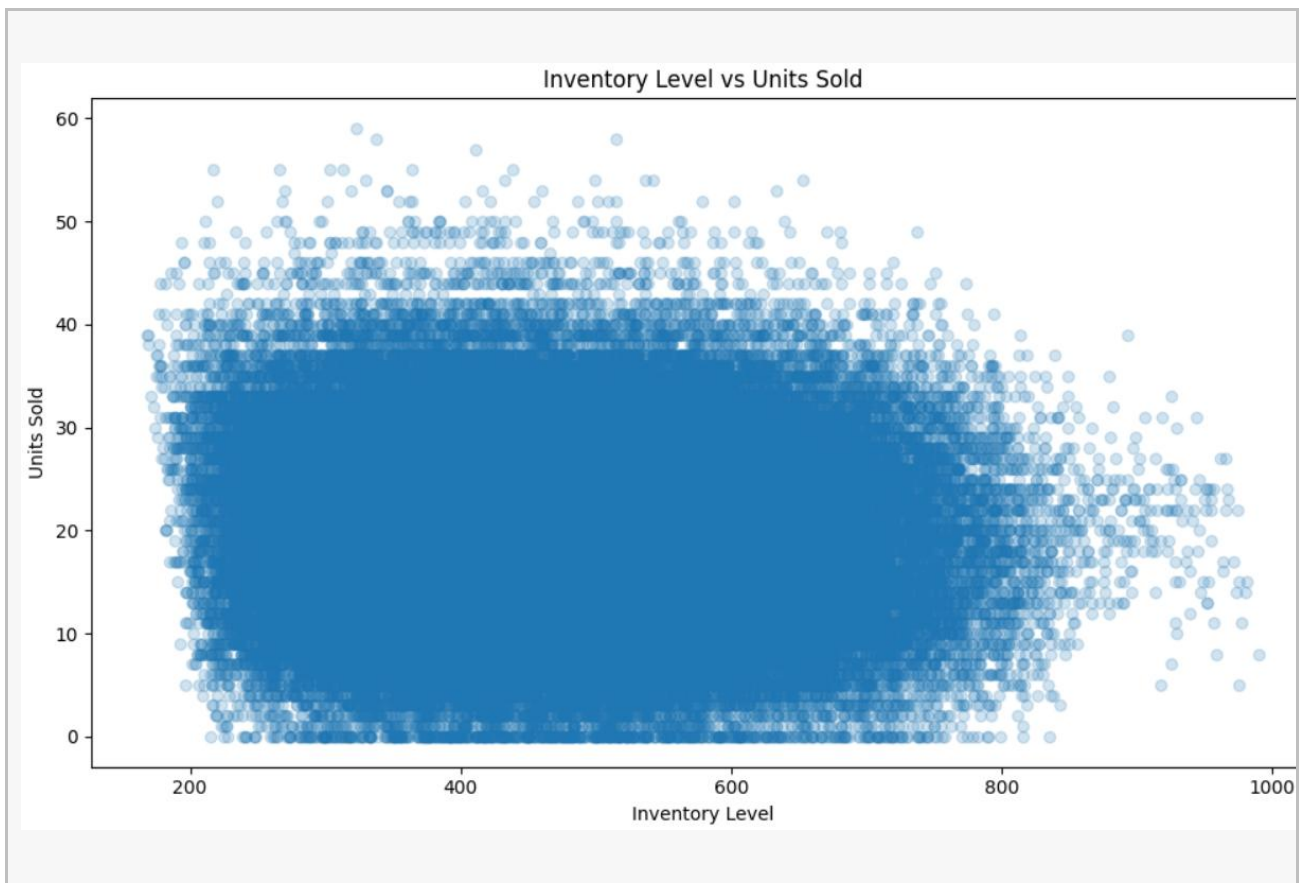


Figure 3. Relationship Between Inventory Level and Units Sold

The scatter plot shows a weak negative relationship between inventory level and units sold. The Pearson correlation coefficient between the two variables is approximately -0.0643 , indicating that the association is very weak despite its negative direction. The observations are widely dispersed across different inventory levels, suggesting that inventory level alone does not explain substantial variation in daily demand. From a modelling perspective, this indicates that additional historical, temporal, promotional, and other supply chain variables are needed to better represent demand patterns. This relationship should be interpreted as an association in the observed data and not as evidence that inventory level causes changes in demand.

8.4 Correlation Analysis

A correlation analysis was performed to examine the linear relationships between Units_Sold and the numerical variables in the dataset. Figure 4 presents the correlation heatmap, providing an overview of the direction and strength of the relationships among the numerical variables.

```
# Visualize the correlation matrix

plt.figure(figsize=(12, 9))

plt.imshow(correlation_matrix, aspect="auto")

plt.colorbar(label="Correlation")

plt.xticks(
    range(len(correlation_matrix.columns)),
    correlation_matrix.columns,
    rotation=90
)

plt.yticks(
    range(len(correlation_matrix.columns)),
    correlation_matrix.columns
)

plt.title("Correlation Matrix of Numerical Variables")

plt.tight_layout()
plt.show()
```

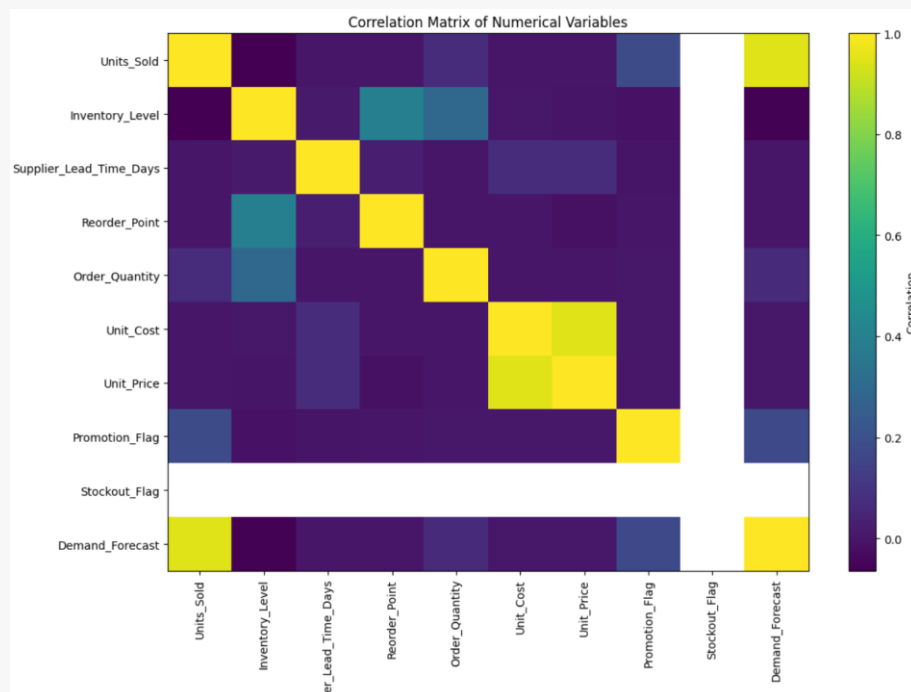


Figure 4. Correlation Heatmap of Numerical Variables

The correlation heatmap shows that most numerical variables have relatively weak linear relationships with Units_Sold. The strongest positive association with demand among the examined variables is observed for Promotion_Flag ($r = 0.1802$), followed by Order_Quantity ($r = 0.0693$). Inventory_Level shows a weak negative relationship with demand ($r = -0.0643$). The remaining numerical variables, including supplier lead time, reorder point, unit cost, and unit price, show correlations close to zero with Units_Sold. These results suggest that no single numerical variable provides a strong linear explanation of demand and support the use of multiple historical, temporal, operational, and commercial features in the forecasting models. Correlation indicates association rather than causation.

8.5 Other Figures

In addition to the main exploratory analyses, the variation in demand across different operational and categorical dimensions was examined to provide a broader understanding of the dataset. Figure 5 presents the average Units_Sold across selected warehouse groups, while additional visualizations such as demand by SKU and region were used to examine whether substantial differences existed across these categories.

```
# Plot average demand by warehouse
```

```
plt.figure(figsize=(9, 6))
```

```
warehouse_demand.plot(kind="bar")
```

```
plt.title("Average Units Sold by Warehouse")
```

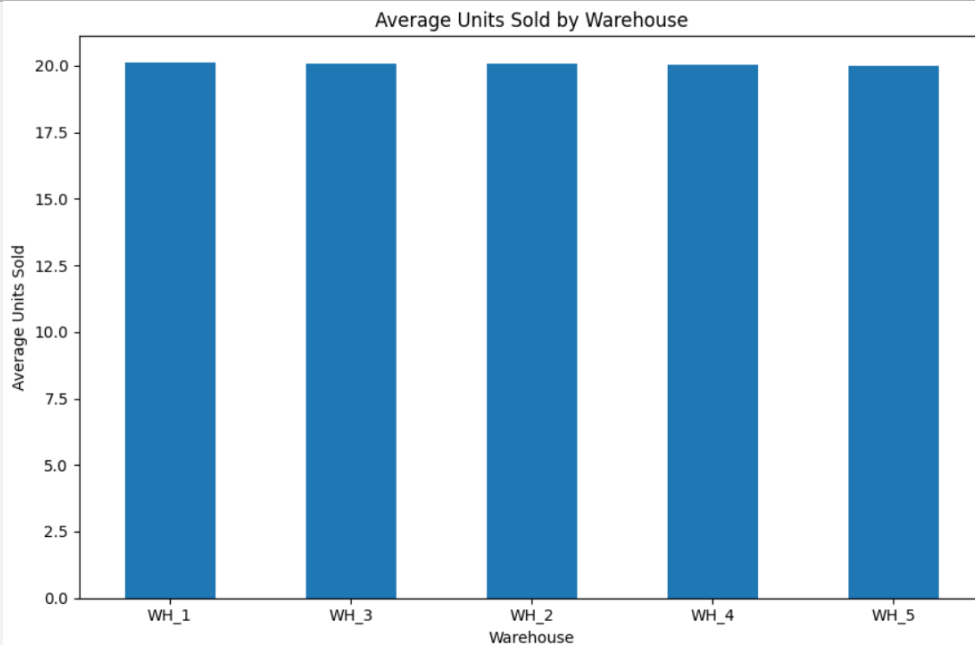
```
plt.xlabel("Warehouse")
```

```
plt.ylabel("Average Units Sold")
```

```
plt.xticks(rotation=0)
```

```
plt.tight_layout()
```

```
plt.show()
```

**Figure 5. Average Units Sold by Warehouse**

The comparison of average units sold across warehouses shows that the average demand levels are relatively similar across the warehouse groups in the dataset. No warehouse demonstrates a substantially different average demand pattern compared with the others. This suggests that warehouse identity alone provides limited separation in average demand within the observed data. Therefore, warehouse information can still be retained as a categorical feature for modelling, but demand forecasting should also consider historical demand, temporal patterns, promotional activity, and other operational variables rather than relying on warehouse-level differences alone.

The additional SKU- and region-level comparisons similarly indicate relatively small differences in average demand across these groups. These observations support the use of multiple features together when developing the forecasting models.

9. STATISTICAL ANALYSIS

Statistical analysis was performed to examine the characteristics of the supply chain variables and to support the relationships identified during exploratory data analysis. Descriptive statistics were used to understand the central tendency and variation of the main numerical variables. The analysis also examined the relationship between promotional activity and demand, the statistical significance of correlations between demand and selected numerical variables, and differences in demand across regions.

The descriptive statistics of the key numerical supply chain variables are presented in Table 4.

Variable	Mean	Standard Deviation	Minimum	Maximum
Units Sold	20.05	9.07	0	59
Inventory Level	471.52	133.49	168	990
Supplier Lead Time Days	7.98	3.91	2	14
Reorder Point	300.07	54.88	201	398
Order Quantity	19.27	82.34	0	499
Unit Cost	12.20	4.57	5.02	19.76
Unit Price	18.26	7.12	6.95	35.10

Table 4. Descriptive Statistics of Key Variables

The descriptive statistics show that Units_Sold has an average of 20.05 units, with values ranging from 0 to 59 units. The standard deviation of 9.07 indicates variation in daily demand. Inventory_Level has an average of 471.52 units and ranges from 168 to 990 units. The average supplier lead time is 7.98 days, while the average reorder point is 300.07 units. Order_Quantity has an average of 19.27 units and a relatively high standard deviation of 82.34, reflecting substantial variation in order quantities. The average unit cost is 12.20, while the average unit price is 18.26.

The statistical analysis also examined whether demand differed between promotional and non-promotional observations. Average demand was approximately 24.9 units during promotional periods, compared with approximately 19.5 units during non-promotional periods. An independent samples t-test produced a t-statistic of 44.6215 and a p-value below 0.001, indicating a statistically significant difference in mean demand between the two groups. This demonstrates an association between promotional activity and observed demand, although the result does not establish a causal relationship.

Pearson correlation significance analysis showed that Promotion_Flag had a positive correlation with Units_Sold ($r = 0.1802$, $p < 0.001$). Order_Quantity also showed a statistically significant positive relationship ($r = 0.0693$, $p < 0.001$), while Inventory_Level showed a statistically significant but very weak negative relationship ($r = -0.0643$, $p < 0.001$). The relationships between Units_Sold and Supplier_Lead_Time_Days ($r = 0.0005$, $p = 0.878$), Reorder_Point ($r = -0.0016$, $p = 0.622$), Unit_Cost ($r = 0.0017$, $p = 0.614$), and Unit_Price ($r = 0.0013$, $p = 0.696$) were not statistically significant. Although some relationships were statistically significant, their small correlation magnitudes indicate that the individual variables have weak linear associations with demand.

A one-way ANOVA was also conducted to examine whether average Units_Sold differed across the four regions. The test produced an F-statistic of 0.4842 and a p-value of 0.6933. Since the p-value is

greater than 0.05, there is insufficient statistical evidence to conclude that mean demand differs significantly across the regions.

Overall, the statistical analysis supports the exploratory findings that promotional activity is associated with higher observed demand, while most individual operational and pricing variables have weak linear relationships with demand. The results support the use of multiple historical, temporal, operational, commercial, and categorical features for next-day demand prediction. The observed statistical relationships represent associations within the dataset and should not be interpreted as evidence of causation.

10. FEATURE ENGINEERING

Feature engineering was performed to transform the original supply chain variables into features that could better represent historical demand patterns and temporal behavior. Since the objective was to predict next-day units sold, lagged demand and rolling averages were created using only previous observations to avoid data leakage. Additional calendar-based features were extracted from the Date variable to capture recurring temporal patterns. The engineered features used in the forecasting models are summarized in Table 5.

Engineered Feature	How It Was Created	Purpose	Role
lag_1	Previous day's Units_Sold	Captures the most recent demand level	Predictor
lag_7	Units_Sold from 7 days earlier	Captures weekly demand patterns	Predictor
lag_14	Units_Sold from 14 days earlier	Captures longer short-term demand patterns	Predictor
rolling_mean_7	Mean of previous 7 days' demand	Smooths daily fluctuations and represents recent demand trend	Predictor
rolling_mean_14	Mean of previous 14 days' demand	Represents a broader recent demand trend	Predictor
day_of_week	Extracted from Date	Captures differences in demand across days of the week	Predictor
month	Extracted from Date	Captures monthly demand patterns	Predictor
quarter	Extracted from Date	Represents broader seasonal periods within the year	Predictor
week_of_year	Extracted from Date	Captures variation across different weeks of the year	Predictor
day_of_year	Extracted from Date	Represents the position of an observation within the year	Predictor
Next_Day_Units_Sold	Next day's Units_Sold for the same SKU-Warehouse combination	Defines the one-day-ahead forecasting target	Target

Table 5. Engineered Features

The lag and rolling features were generated using historical Units_Sold values for the same SKU-Warehouse combination. A one-period shift was applied before calculating the rolling averages so that information from the prediction day or future observations was not included. The temporal features were derived from the Date variable to represent weekly and yearly time patterns. Together, these engineered

variables provide the models with information about recent demand behavior and temporal variation that is not directly available from the original raw variables.

11. MODEL DEVELOPMENT

11.1 Problem Formulation

The problem was formulated as a regression task because the objective is to predict the numerical value of `Next_Day_Units_Sold`. The target represents the actual number of units sold on the following day for the same SKU-Warehouse combination.

The input features include historical demand features (`lag_1`, `lag_7`, `lag_14`, `rolling_mean_7`, and `rolling_mean_14`), temporal features (`day_of_week`, `month`, `quarter`, `week_of_year`, and `day_of_year`), categorical information (`SKU_ID`, `Warehouse_ID`, `Supplier_ID`, and `Region`), operational and inventory variables (`Inventory_Level`, `Supplier_Lead_Time_Days`, `Reorder_Point`, and `Order_Quantity`), and commercial variables (`Unit_Cost`, `Unit_Price`, and `Promotion_Flag`).

The regression formulation is appropriate because demand is measured as a continuous numerical quantity and the objective is to estimate future demand rather than assign observations to predefined categories. A one-day-ahead forecasting horizon was selected because next-day demand estimates can support short-term inventory and replenishment planning.

11.2 Model Selection

Four approaches were evaluated to establish a comparison between a simple forecasting approach and machine learning regression models.

Persistence Baseline: The previous day's demand (`lag_1`) was used as a simple baseline. This provides a reference point for determining whether the machine learning models provide useful predictive improvement over using the most recent observed demand.

Linear Regression: Linear Regression was selected as a simple and interpretable regression model. It provides a useful baseline for understanding how the engineered historical, temporal, categorical, operational, and commercial features relate to next-day demand.

Decision Tree Regressor: A Decision Tree Regressor was included because it can capture nonlinear relationships and interactions between features without requiring feature scaling.

Random Forest Regressor: Random Forest was selected as an ensemble extension of the decision-tree approach. It combines multiple decision trees and can model more complex nonlinear relationships while generally providing greater robustness than a single decision tree.

The models were evaluated using the same chronological train-test split and the same preprocessing approach to provide a consistent comparison. Linear Regression achieved the strongest test performance among the evaluated models, while the tree-based models showed substantial differences between training and testing performance, indicating overfitting.

11.3 Training the Model

The models were trained using the engineered feature set and the chronological training data. Categorical variables were transformed using One-Hot Encoding through a `ColumnTransformer`, while

numerical variables were passed through without scaling. The preprocessing pipeline was fitted using the training data and subsequently applied to the test data.

The main training procedure for the final Random Forest model was:

```
from sklearn.ensemble import RandomForestRegressor

# Create the Random Forest model
random_forest_model = RandomForestRegressor(
    n_estimators=100,
    random_state=42,
    n_jobs=-1
)

# Train the model
random_forest_model.fit(
    X_train_processed,
    y_train
)

print("Random Forest model trained successfully.")
```

The Random Forest Regressor was configured with 100 trees (`n_estimators=100`) and a fixed random state of 42 for reproducibility. Parallel processing was enabled using `n_jobs=-1`. The remaining hyperparameters were kept at their default values.

The same training dataset and preprocessing representation were used for Linear Regression and Decision Tree Regression. Model performance was subsequently evaluated on the unseen chronological test set using MAE, RMSE, and R^2 .

12. MODEL EVALUATION

The trained regression models were evaluated on the unseen chronological test set using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 (coefficient of determination). These metrics were selected because they measure prediction error and the ability of the models to explain variation in next-day demand. Lower MAE and RMSE indicate smaller prediction errors, while a higher R^2 indicates better explanatory performance.

12.1 Regression Evaluation

MAE represents the average absolute difference between the actual and predicted demand values and is expressed in units sold. RMSE measures the square root of the average squared prediction error and gives greater weight to larger errors. R^2 indicates how much of the variation in the target variable is explained by the model, with values closer to 1 indicating stronger explanatory performance.

The test-set performance of the evaluated models is presented in Table 6.

Model	Test MAE	Test RMSE	Test R^2
Persistence Baseline	5.8938	7.4303	-0.5179
Linear Regression	5.4290	6.8054	-0.2733
Decision Tree Regressor	8.2031	10.1523	-1.8337
Random Forest Regressor	5.9319	7.4302	-0.5178

Table 6. Regression Model Performance

The results show that Linear Regression produced the lowest test MAE (5.4290) and lowest test RMSE (6.8054) among the evaluated models. Its test R^2 was -0.2733 . The Persistence Baseline and Random Forest produced similar test performance, while the Decision Tree produced the largest prediction errors.

The negative R^2 values indicate that the evaluated models did not explain the variation in unseen next-day demand sufficiently well relative to a constant-mean prediction reference. Therefore, although Linear Regression had the strongest test performance among the evaluated approaches based on the reported metrics, the overall predictive performance remained limited.

12.2 Overfitting and Underfitting

The difference between training and testing performance was examined to assess model generalization. The Linear Regression model achieved a training MAE of 4.4323 and a test MAE of 5.4290, showing a relatively smaller difference between training and testing error compared with the tree-based models.

The Decision Tree Regressor achieved a training MAE of 0.0000 and training R^2 of 1.0000, but its test MAE increased to 8.2031 and test R^2 decreased to -1.8337 . This large difference between training and test performance indicates substantial overfitting, where the model learned the training observations very closely but did not generalize well to unseen observations.

The Random Forest model also showed a considerable difference between training and test performance. Its training MAE was 1.6384, compared with a test MAE of 5.9319, while training R^2 was 0.9497 compared with test R^2 of -0.5178 . This indicates that the Random Forest model also overfit the training data to a considerable extent.

Overall, Linear Regression showed the smallest train-test MAE gap among the evaluated models and therefore demonstrated comparatively more stable generalization. However, its negative test R^2 indicates that the forecasting performance was still limited on the unseen test period.

13. RESULTS AND FINDINGS

The analysis produced several important findings related to demand patterns, supply chain variables, promotional activity, and next-day demand forecasting. The key findings are summarized below.

1. Demand shows substantial variation across daily observations :

The distribution of Units_Sold shows that daily demand ranges from 0 to 59 units, with an average of approximately 20.05 units. The variation in observed demand indicates that demand is not constant across SKU-Warehouse-Day observations. This supports the need to consider historical demand and temporal information when developing a next-day forecasting model. (Figure 1; Table 4)

2. Promotional periods are associated with higher observed demand :

Average demand was approximately 24.9 units during promotional periods, compared with approximately 19.5 units when no promotion was active. An independent samples t-test produced a t-statistic of 44.6215 with $p < 0.001$, indicating a statistically significant difference in mean demand between the two groups. This suggests that promotional activity is an important variable to consider in demand and inventory planning, although the result represents an association rather than proof of causation. (Figure 2; Section 9)

3. Most individual operational and pricing variables have weak linear relationships with demand :

The correlation analysis found relatively weak relationships between Units_Sold and most numerical supply chain variables. Inventory_Level had a weak negative correlation ($r = -0.0643$), while Order_Quantity had a weak positive correlation ($r = 0.0693$). Unit_Price, Unit_Cost, Reorder_Point, and Supplier_Lead_Time_Days showed correlations close to zero. This indicates that demand cannot be adequately represented using a single operational or pricing variable and supports the use of multiple features together. (Figure 3; Figure 4; Section 9)

4. Linear Regression produced the strongest test performance among the evaluated models, but predictive performance remained limited :

The Linear Regression model achieved a test MAE of 5.4290, RMSE of 6.8054, and R^2 of -0.2733 , giving it the lowest test MAE and RMSE among the evaluated approaches. However, the negative test R^2 indicates that the model did not explain the variation in unseen next-day demand sufficiently well. Therefore, the results provide a useful baseline forecasting framework but also indicate that additional features or improved modelling approaches are needed for stronger predictive performance. (Table 6; Section 12.1)

5. Tree-based models showed substantial overfitting between training and test data.:

The Decision Tree achieved a training MAE of 0.0000 and training R^2 of 1.0000, but its test MAE increased to 8.2031 and test R^2 decreased to -1.8337 . The Random Forest also showed a substantial training-test difference, with training MAE of 1.6384 compared with test MAE of 5.9319. These differences indicate that the tree-based models fitted the training data closely but generalized less effectively to the unseen test period. (Table 6; Section 12.2)

14. DISCUSSION

The analysis provides insights into the demand patterns and forecasting challenges present in the supply chain dataset. The observed variation in daily Units_Sold indicates that demand changes across SKU-Warehouse-Day observations, supporting the use of historical and temporal information for next-day demand forecasting. The engineered lag and rolling features were therefore used to represent recent demand behavior and temporal patterns in the forecasting models.

One of the clearest findings was the difference in observed demand between promotional and non-promotional periods. Promotional observations had higher average demand, and the statistical test indicated a significant difference between the two groups. From a supply chain perspective, this relationship suggests that promotional periods should be considered when planning inventory and replenishment. However, the analysis identifies an association between promotion and demand and does not establish that promotional activity alone caused the increase.

The correlation analysis showed that most individual operational and pricing variables had weak linear relationships with demand. In particular, inventory level, order quantity, supplier lead time, reorder point, unit cost, and unit price did not individually provide a strong linear explanation of demand. This suggests that demand behavior is influenced by a combination of factors rather than being adequately represented by a single supply chain variable. The inclusion of historical demand and temporal features therefore provides a more appropriate representation for the forecasting problem.

The model evaluation also highlights the difficulty of forecasting unseen next-day demand using the available variables. Linear Regression produced the strongest test performance among the evaluated models, but its negative test R^2 indicates that the model's predictive capability remained limited. The Decision Tree and Random Forest models achieved substantially stronger training performance but

showed poorer generalization on the test period. This difference suggests that the tree-based models captured patterns specific to the training data that did not transfer effectively to unseen observations.

From a practical perspective, the findings suggest that historical demand patterns and promotional information can be useful inputs for short-term supply chain planning, but the current dataset does not provide sufficient predictive strength for relying on the evaluated models as a standalone forecasting system. More extensive historical data, additional demand-related variables, and alternative forecasting approaches could be explored to improve predictive performance.

Overall, the results demonstrate that the provided supply chain data can be used to identify demand patterns and develop a baseline next-day forecasting framework. At the same time, the limited test performance and observed overfitting indicate that the findings should be interpreted within the scope of the available dataset and should not be treated as evidence of causal relationships.

15. LIMITATIONS

- **Limited time period:** The dataset covers approximately one year of observations. This limits the ability to identify and validate recurring annual seasonal patterns and reduces the extent to which the findings can be generalized to longer time periods.
- **Limited predictive performance:** Although Linear Regression achieved the strongest test performance among the evaluated models, its test R^2 was **-0.2733**, indicating limited predictive performance on unseen next-day demand. The results therefore represent a baseline forecasting framework rather than a highly accurate forecasting system.
- **Model overfitting:** The Decision Tree and Random Forest models showed substantially stronger training performance than testing performance. This indicates that the models captured patterns in the training data that did not generalize effectively to the unseen test period.
- **Limited variation in some variables:** Stockout_Flag contained only one value and was therefore excluded from modelling. In addition, average demand was relatively similar across SKU, warehouse, and regional groups, limiting the ability of these categorical variables to provide strong separation in demand.
- **Potential information leakage in the original dataset:** The original Demand_Forecast variable had a very strong relationship with actual Units_Sold. It was therefore excluded from the predictive feature set to avoid using information that could make the forecasting results misleading.
- **Dataset-specific findings:** The conclusions are based on the characteristics of the provided P3 Supply Chain Dataset. Therefore, the observed relationships and model performance may not directly generalize to other supply chains, products, warehouses, or time periods.
- **Limited causal interpretation:** The statistical analysis identifies associations between variables, such as promotional activity and demand, but does not establish causal relationships. Additional experimental or longitudinal analysis would be required to determine causal effects.

16. CONCLUSION

This project developed a data-driven approach for analyzing supply chain demand and forecasting next-day units sold using historical demand, temporal, inventory, operational, pricing, promotional, and categorical information. The analysis identified meaningful variation in demand and a significant association between promotional activity and observed demand, while most individual operational and pricing variables showed weak linear relationships with demand.

Four approaches were evaluated using a chronological train-test framework. Linear Regression achieved the strongest test performance among the evaluated models, with a test MAE of 5.4290, RMSE of 6.8054, and R^2 of -0.2733 . However, the negative test R^2 and the overfitting observed in the tree-based models demonstrate that the available data and selected models provided limited predictive performance on unseen demand.

Overall, the project demonstrates how data preprocessing, exploratory analysis, statistical analysis, feature engineering, and machine learning can be combined to support supply chain demand analysis. The resulting framework provides a baseline for data-driven demand planning and inventory management, while highlighting the need for richer data and improved forecasting methods for more reliable future predictions.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

- **Incorporate promotional activity into demand and inventory planning:**
The analysis found higher observed demand during promotional periods, with average demand of approximately 24.9 units compared with 19.5 units during non-promotional periods. Promotional information should therefore be considered when preparing short-term demand and replenishment plans.
- **Use recent historical demand for short-term planning:**
The feature engineering and forecasting approach demonstrated the importance of representing recent demand through lagged demand and rolling-average features. Inventory planning can incorporate recent sales history alongside operational and promotional information when estimating near-term demand.
- **Use model predictions as supporting information rather than as the sole basis for inventory decisions:**
Since the Linear Regression model achieved a test MAE of 5.4290 but a negative test R^2 of -0.2733 , the current forecasting model should be treated as a baseline analytical tool. Inventory and replenishment decisions should also consider operational knowledge and other available supply chain information.
- **Monitor model performance on new demand data:**
The tree-based models showed substantial differences between training and testing performance. Any future forecasting implementation should therefore regularly evaluate prediction errors on newly observed data and reassess the model when performance changes.

17.2 Future Work

Future work can extend the study by incorporating a longer historical time period to better capture recurring seasonal and yearly demand patterns. Additional variables such as holidays, special events, discounts, customer demand indicators, supplier disruptions, and external market factors could also provide more information for forecasting.

Alternative forecasting and machine learning methods, including more advanced time-series and ensemble approaches, can be evaluated against the current baseline models. Future studies could also use time-series cross-validation and additional forecasting horizons such as 7-day demand prediction to assess model stability more rigorously.

The forecasting framework could eventually be integrated into an inventory decision-support system that combines demand predictions with inventory levels, reorder points, supplier lead times, and order quantities to support replenishment planning. Further validation using real-world supply chain data would be required before such a system could be used operationally.

18. REFERENCES

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APPENDIX A. CODE

A.1 Data Loading and Inspection

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("P_3_supply_chain_dataset1.csv")

print("Dataset Shape:", df.shape)
print(df.info())
print(df.isnull().sum())
print("Duplicate Rows:", df.duplicated().sum())
```

A.2 Data Cleaning

```
df["Date"] = pd.to_datetime(df["Date"])
```



```
df_processed = df.copy()

df_processed = df_processed.drop(
    columns=["Stockout_Flag", "Demand_Forecast"]
)

df_processed = df_processed.sort_values(
    by=["SKU_ID", "Warehouse_ID", "Date"]
).reset_index(drop=True)
```

A.3 Preprocessing

```
df_processed["Next_Day_Units_Sold"] = (
    df_processed.groupby(
        ["SKU_ID", "Warehouse_ID"]
    )["Units_Sold"].shift(-1)
)

grouped_sales = df_processed.groupby(
    ["SKU_ID", "Warehouse_ID"]
)["Units_Sold"]

df_processed["lag_1"] = grouped_sales.shift(1)
df_processed["lag_7"] = grouped_sales.shift(7)
df_processed["lag_14"] = grouped_sales.shift(14)

df_processed["rolling_mean_7"] = (
    df_processed.groupby(
        ["SKU_ID", "Warehouse_ID"]
    )["Units_Sold"]
    .transform(lambda x: x.shift(1).rolling(7).mean())
)

df_processed["rolling_mean_14"] = (
    df_processed.groupby(
        ["SKU_ID", "Warehouse_ID"]
    )["Units_Sold"]
    .transform(lambda x: x.shift(1).rolling(14).mean())
)

df_processed["day_of_week"] = df_processed["Date"].dt.dayofweek
df_processed["month"] = df_processed["Date"].dt.month
df_processed["quarter"] = df_processed["Date"].dt.quarter
df_processed["week_of_year"] = (
    df_processed["Date"].dt.isocalendar().week.astype(int)
)
df_processed["day_of_year"] = df_processed["Date"].dt.dayofyear
```

A.4 EDA and Visualization

```
# Distribution of Units Sold
plt.figure(figsize=(10, 6))
plt.hist(df["Units_Sold"], bins=30)
plt.title("Distribution of Units Sold")
plt.xlabel("Units Sold")
plt.ylabel("Frequency")
plt.show()

# Promotion vs Demand
promotion_demand = df.groupby(
    "Promotion_Flag"
)["Units_Sold"].mean()
```

```
promotion_demand.plot(kind="bar")
plt.title("Average Units Sold by Promotion Status")
plt.xlabel("Promotion Flag")
plt.ylabel("Average Units Sold")
plt.show()

# Correlation Analysis
correlation_matrix = df.select_dtypes(
    include=np.number
).corr()

plt.figure(figsize=(12, 9))
plt.imshow(correlation_matrix, aspect="auto")
plt.colorbar()
plt.title("Correlation Matrix of Numerical Variables")
plt.show()
```

A.5 Model Training and Evaluation

```
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Linear Regression
linear_model = LinearRegression()
linear_model.fit(X_train_processed, y_train)

# Decision Tree
decision_tree_model = DecisionTreeRegressor(random_state=42)
decision_tree_model.fit(X_train_processed, y_train)

# Random Forest
random_forest_model = RandomForestRegressor(
    n_estimators=100,
    random_state=42,
    n_jobs=-1
)
random_forest_model.fit(X_train_processed, y_train)

# Predictions
linear_pred = linear_model.predict(X_test_processed)
tree_pred = decision_tree_model.predict(X_test_processed)
rf_pred = random_forest_model.predict(X_test_processed)

# Evaluation
for name, pred in [
    ("Linear Regression", linear_pred),
    ("Decision Tree", tree_pred),
    ("Random Forest", rf_pred)
]:
    print(name)
    print("MAE:", mean_absolute_error(y_test, pred))
    print("RMSE:", np.sqrt(mean_squared_error(y_test, pred)))
    print("R2:", r2_score(y_test, pred))
```

APPENDIX B. ADDITIONAL FIGURES AND TABLES

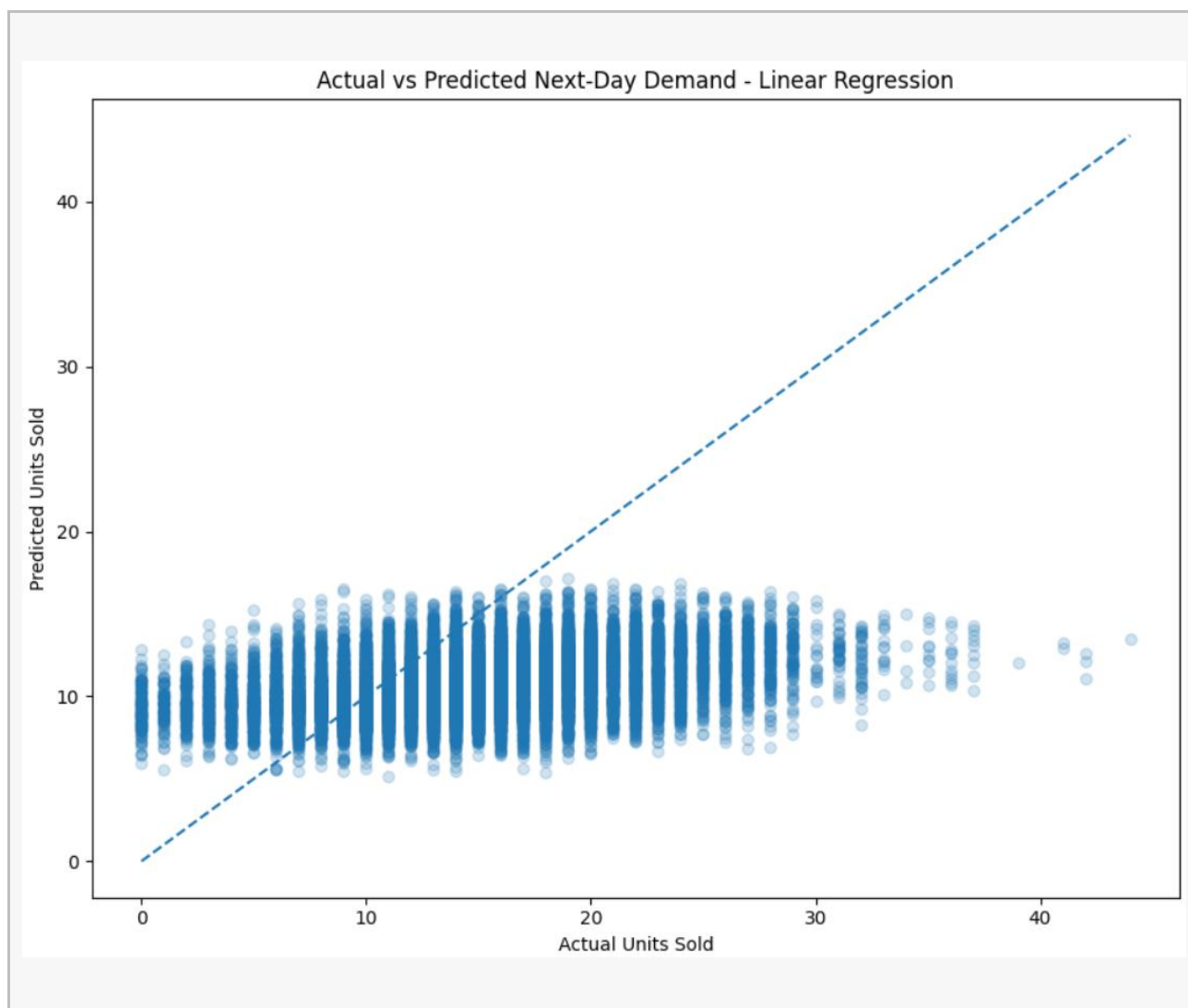


Figure 6. Actual vs Predicted Next-Day Demand – Linear Regression

Table 7 shows a clear difference between training and testing performance, particularly for the Decision Tree and Random Forest models. The Linear Regression model shows a comparatively smaller difference between training and testing MAE, while the tree-based models show larger performance gaps, indicating weaker generalization to the unseen test data.

Model	Training MAE	Testing MAE	Training R ²	Testing R ²
Linear Regression	4.4323	5.4290	0.6377	-0.2733
Decision Tree	0.0000	8.2031	1.0000	-1.8337
Random Forest	1.6384	5.9319	0.9497	-0.5178

Table 7. Training and Testing Performance Comparison